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CAPACITOR, MEMORY AND MANUFACTURING METHOD OF MEMORY

Abstract

A capacitor includes a first electrode layer, a second electrode layer; and a strontium titanate dielectric layer formed between the first electrode layer and the second electrode layer, where in a direction from the center of the strontium titanate dielectric layer to two opposite sides of the strontium titanate dielectric layer, ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually, one side of the two opposite sides in the strontium titanate dielectric layer is close to the first electrode layer, and the other side of the two opposite sides in the strontium titanate dielectric layer is close to the second electrode layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/CN2024/094149, filed on May 20, 2024, which claims priority to Chinese Patent Application No. 202310924147.9, filed on Jul. 25, 2023, and Chinese Patent Application No. 202321986642.4, filed on Jul. 25, 2023, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD
[0002] The present application pertains to the field of chips, and particularly relates to a capacitor, a memory and a manufacturing method of the memory.
BACKGROUND
[0003] With the development of chip industry, there are increasingly high requirements on Dynamic Random Access Memory (DRAM) chips with higher storage density in the market. How to realize the higher storage density of the DRAM chip is a significant research topic in many chip industries.
[0004] As far as the DRAM chip including 1T1C (one transistor+1 capacitor) is concerned, the capacitor is an important factor which restrains critical size of the DRAM chip. How to improve the storage density of the DRAM on the premise of reducing the critical size to guarantee that the DRAM chip has an enough signal resolution, i.e., how to guarantee a higher dielectric constant (K value for short) and a lower leakage density of the capacitor becomes a problem needed to be solved urgently.
SUMMARY
[0005] There is provided a capacitor, a memory and a method manufacturing of the memory according to embodiments of the present disclosure. The technical solution is as below:
[0006] A first aspect of the present application provides a capacitor, which includes: [0007] a first electrode layer; [0008] a second electrode layer; and [0009] a strontium titanate dielectric layer formed between the first electrode layer and the second electrode layer; [0010] where in a direction from a center of the strontium titanate dielectric layer to two opposite sides of the strontium titanate dielectric layer, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually; [0011] one side of the two opposite sides in the strontium titanate dielectric layer is close to the first electrode layer and the other side of the two opposite sides in the strontium titanate dielectric layer is close to the second electrode layer.
[0012] A second aspect of the present application provides a memory, which includes a substrate and any one capacitor above, the capacitor being formed on the substrate.
[0013] A third aspect of the present application provides a manufacturing method of a memory, which may include: [0014] providing a substrate; [0015] forming a first electrode layer on the substrate; [0016] forming a strontium titanate dielectric layer at a side of the first electrode layer away from the substrate, where in a direction from a center of the strontium titanate dielectric layer to two opposite sides of the strontium titanate dielectric layer, ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually, one side of the two opposite sides in the strontium titanate dielectric layer is close to the first electrode layer and the other side of the two opposite sides in the strontium titanate dielectric layer is away from the first electrode layer; and [0017]

forming a second electrode layer at a side of the strontium titanate dielectric layer away from the first electrode layer to form a capacitor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments consistent with the present application and, together with the description, serve to explain the principles of the present application. It is apparent that the drawings described below are merely some embodiments of the present application, and those of ordinary skill in the art further can obtain other drawings according to those drawings without making creative efforts.

[0019] FIGS. **1-5** show schematic structural diagrams of memories corresponding to various solutions mentioned in Embodiment I of the present application.

[0020] FIG. **6** shows a three-dimensional schematic structural diagram of the memories mentioned in Embodiment I of the present application.

[0021] FIG. **7** shows a schematic structural diagram of a capacitor in a solution mentioned in Embodiment I of the present application.

[0022] FIGS. **8-10** show schematic diagrams of corresponding changes when the ratios of Sr/(Sr+Ti) in a strontium titanate dielectric layer are designed at different gradients in Embodiment I of the present application.

[0023] FIG. **11** shows a schematic structural diagram of a capacitor in another solution mentioned in Embodiment I of the present application.

[0024] FIGS. **12-14** show schematic structural diagrams of capacitors in various corresponding solutions in Embodiment II of the present application.

[0025] FIGS. **15-25** show schematic structural diagrams corresponding to steps in a manufacturing method of a memory provided in Embodiment III of the present application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0026] The exemplary implementations will be described more comprehensively with reference to drawings. However, the exemplary embodiments can be implemented in various forms and shall not be construed as examples to the described embodiments. On the contrary, the embodiments are provided to make the present application more comprehensive and integral and the concept of the exemplary embodiment is comprehensively transferred to those skilled in the art.

[0027] In addition, features, structures or characteristics described herein may be combined in one or more embodiments freely as appropriate. In the description below, many specific details are provided, so that the embodiments of the present application are fully understood. However, those skilled in the art will realize that the technical solution of the present application can be practiced without one or more of specific details or other methods, components, devices, steps and the like can be used. In other cases, known methods, devices, implementations or operations are not illustrated or described in detail to prevent obscuring all aspects of the present application.

[0028] The present application will be further described in detail below in combination with the drawings and the specific embodiments. It is to be noted herein that technical features involved in the described embodiments of the present application can be combined with one another as long as they do not conflict with each other. The embodiments described with reference to the drawings below are exemplary, and are merely used for explaining the present application and are not construed as a limitation to the present application.

Embodiment I

[0029] The embodiment of the present application provides a memory, which can be a DRAM, for example, a three-dimensional DRAM, i.e., a memory formed by stacking multiple layers of DRAM. But it shall be appreciated that the memory in the embodiment is not limited to DRAM but

may also be a memory of another type, which depends on circumstances.
[0030] As shown in FIG. **1**, the memory in the embodiment may include a substrate **10** and a plurality of storage units formed on the substrate **10**, the storage units can include one capacitor **11**, and the capacitor **11** is formed on the substrate **10**.
[0031] The capacitor **11** may include a first electrode layer **110**, a dielectric layer **111** and a second electrode layer **112** formed in a stacked manner in sequence on the substrate **10**, that is to say, the dielectric layer **111** is formed between the first electrode layer **110** and the second electrode layer **112**.
[0032] For example, the capacitor **11** may be a Metal-Isolation-Metal (MIM) structure, i.e., the first electrode layer **110** and the second electrode layer **112** each includes a metal or a metal compound and has a conducting property. The dielectric layer **111** is an insulator for isolating the first electrode layer **110** and the second electrode layer **112**.
[0033] As shown in FIG. **2**, the substrate **10** may include a semiconductor base **100**, for example, a silicon base, but is not limited thereto. The substrate may also be a base formed by another semiconductor material (for example, silicon carbide, a gallium compound and a germanium compound). In order to isolate influence of the semiconductor base **100** on the MIM structure, the substrate **10** may further include forming an insulation isolation layer **101** between the first electrode layer **110** and the semiconductor base **100**, that is to say, prior to forming the first electrode layer **110** on the semiconductor base **100**, forming the insulation isolation layer **101** on the semiconductor base **100** first, and then forming the first electrode layer **110** to insulate the semiconductor base **100** and the first electrode layer **110** with the insulation isolation layer **101**.
[0034] For example, the insulation isolation layer **101** may be a SiO.sub.2 (silicon dioxide) material, but is not limited thereto. The insulation isolation layer may also be another insulating material, which depends on circumstances.
[0035] To measure the electric property of the dielectric layer **111** in the capacitor **11**, in the process of manufacturing the capacitor **11** on the substrate **10**, a measurement electrode for measuring the capacitor **11** may also be manufactured, and a specific relation between the capacitor **11** and the measurement electrode is as follows.
[0036] As shown in FIG. **3**, in the capacitor **11**, orthographic projections of the second electrode layer **112** and the dielectric layer **111** on the substrate **10** are fall with a middle area of an orthographic projection of the first electrode layer **110** on the substrate **10**, that is to say, an edge area of the first electrode layer **110** is not covered by the second electrode layer **112** and the dielectric layer **111**, so that the subsequently manufactured measurement electrode is in contact with the first electrode layer **110** conveniently, where the orthographic projection of the second electrode layer **112** on the first electrode layer **110** is overlapped with the orthographic projection of the dielectric layer **111** on the first electrode layer **110** to guarantee the storage capacity of the capacitor **11**.
[0037] As shown in FIG. **4**, after forming the capacitor **11** on the substrate **10**, the memory further includes an insulation dielectric layer **12** formed on the substrate **10**, an etch stop layer **13** and a first measurement electrode **14***a* and a second measurement electrode **14***b* arranged at intervals.
[0038] The insulation dielectric layer **12** covers the edge area of the first electrode layer **110** and the second electrode layer **112**, the etch stop layer **13** is formed at a side of the insulation dielectric layer **12** away from the substrate **10**, and the etch stop layer **13** fully covers the insulation dielectric layer **12**. The first measurement electrode **14***a* and the second measurement electrode **14***b* each includes a via hole conduction portion **140** and a measurement conduction portion **141** connected with each other, the measurement conduction portion **141** of the first measurement electrode **14***a* and the measurement conduction portion of the second measurement electrode **14***b* are formed at a side of the etch stop layer **13** away from the substrate **10** for being connected to an external measurement device which measures the performance of the capacitor **11**.
[0039] An orthographic projection of the via hole conduction portion **140** of the first measurement

electrode **14***a* on the substrate **10** fall with an orthographic projection of the edge area of the first electrode layer **110** on the substrate **10**, the via hole conduction portion **140** of the first measurement electrode **14***a* penetrates through the etch stop layer **13** and the insulation dielectric layer **12** in sequence and is in contact with the edge area of the first electrode layer **110**, so that one pole of the measurement device is in contact with the first electrode layer **110** through the first measurement electrode **14***a*; an orthographic projection of the via hole conduction portion **140** of the second measurement electrode **14***b* on the substrate **10** falls within an orthographic projection of the second electrode layer **112** on the substrate **10**, the via hole conduction portion **140** of the second measurement electrode **14***b* penetrates through the etch stop layer **13** and the insulation dielectric layer **12** in sequence and is in contact with the edge area of the second electrode layer **112**, so that the other pole of the measurement device is in contact with the second electrode layer **112** through the first measurement electrode **14***b*. In the embodiment, a material of the first measurement electrode **14***a* and the second measurement electrode **14***b* may be TiN (titanium nitride) and the like, but is not limited thereto. The material may also be another conducting material, which depends on circumstances.

[0040] Since the dielectric layer **111** in the capacitor **11** is very thin, usually several nanometers, electric leakage at the edge of the dielectric layer **111** is large. Electric leakage at the edge does not pertain to the intrinsic property of the material, and will interfere with a measured effect. On this basis, as shown in FIG. **5**, the memory in the embodiment may further include an interlevel dielectric layer **15** to open the distance between the edge of the second electrode layer **112** and the first electrode layer **110** larger.

[0041] Specifically, as shown in FIG. **5**, the interlevel dielectric layer **15** may be formed between the first electrode layer **110** and the insulation dielectric layer **12**. It shall be appreciated that the via hole conduction portion **140** of the abovementioned first measurement electrode **14***a* further penetrates through the interlevel dielectric layer **15** to be in contact with the edge area of the first electrode layer **110** while penetrating through the etch stop layer **13** and the insulation dielectric layer **12**. Moreover, the interlevel dielectric layer **15** further has a through hole **150** to expose part of the first electrode layer **110**.

[0042] As shown in FIG. **5**, the dielectric layer **111** has a middle portion and an edge portion arranged around the middle portion. The middle portion of the dielectric layer **111** is formed in the through hole **150** and is in contact with the first electrode layer **110**. The edge portion of the dielectric layer **111** is in overlap joint to a surface of the interlevel dielectric layer **15** away from the first electrode layer **110** to open the distance between the edge of the second electrode layer **112** and the first electrode layer **110** larger, and the orthographic projection of the via hole conduction portion **140** of the second measurement electrode **14***b* on the substrate **10** falls within the orthographic projection of the through hole **150** in the substrate **10**, that is to say, the orthographic projection of the via hole conduction portion **140** of the second measurement electrode **14***b* on the substrate **10** falls within an orthographic projection of the middle portion of the dielectric layer **111** on the substrate **10** to accurately measure the K value and electric leakage in the middle of the dielectric layer **111**.

[0043] For example, as shown in FIGS. **4** and **5**, a plurality of via hole conduction portions **140** are arranged in the first measurement electrode **14***a* and the second measurement electrode **14***b* and the via hole conduction portions are arranged around an axis of the through hole **150** at an equal interval. Thus, in the measuring process, the electric field is distributed more uniformly, and the measuring effect is more accurate.

[0044] It is to be noted that when the plurality of via hole conduction portions **140** are arranged in the first measurement electrode **14***a* and the second measurement electrode **14***b*, there may be only one measurement conduction portion **141** in the first measurement electrode **14***a* and the second measurement electrode **14***b*, but is not limited thereto. The plurality of measurement conduction portions may be arranged and the measurement conduction portions are arranged around the axis of the through hole **150** at equal intervals. The measurement conduction portions may be in one-to-one correspondence connection to the via hole conduction portions **140** and one measurement conduction portion may correspond to more via hole conduction portions.

[0045] The first measurement electrode **14***a* and the second measurement electrode **14***b* mentioned in the embodiment are not limited to a finally produced memory. After a test is completed, in a process of cutting dies, they are fully cut or partially cut and the like, which depends on circumstances but does not affect end use of the product.

[0046] In the embodiment, in combination with FIG. **6**, besides the capacitor **11**, the storage unit may further include a transistor **16**, a word line **17** and a bit line **18**. The transistor **16**, the word line **17** and the bit line **18** all may be formed on the substrate **10**. The grid electrode of the transistor **16** may be connected to the word line **17**, one of source and drain electrodes of the transistor **16** are connected to the bit line **18**, and the other one is connected to the capacitor **11**; specifically, it may be connected to the first electrode layer **110** of the capacitor **11** to realize a read-write operation of data, that is to say, the storage unit may be of a 1T1C structure.

[0047] As far as the memory including the 1T1C (one transistor **16**+one capacitor **11**) structure is concerned, the capacitor **11** is an important factor which restrains critical size of the memory. How to improve the storage density of the DRAM on the premise of reducing the critical size to guarantee that the DRAM chip has an enough signal resolution becomes a problem needed to be solved urgently.

[0048] When the memory is the three-dimensional DRAM, as shown in FIG. **6**, the capacitor **11** may be transversely placed, so that the storage density of the capacitor may be effectively increased by reducing the transverse length of the capacitor **11**. Besides, the first electrode layer **110**, the second electrode layer **112** and the dielectric layer **111** of the capacitor **11** may further be adjusted, so that the capacitor **11** has a higher dielectric constant (K value for short), i.e., the storage density is increased, and the electric leakage is reduced. Besides the three-dimensional DRAM, this mode may further be applicable to memories including a capacitor structure. The structure of the capacitor **11** in embodiments of the present application will be descried in detail below in conjunction with the accompanying drawings.

[0049] In the embodiment, as shown in FIG. **7**, the dielectric layer **111** of the capacitor may include a strontium titanate dielectric layer **1110** (STO dielectric layer for short), the strontium titanate dielectric layer **1110** including strontium (Sr) and titanium (Ti).

[0050] According to research, when the strontium titanate dielectric layer **1110** is integrally designed with high Sr content, i.e., the Sr content in the overall design of the strontium titanate dielectric layer **1110** is greater than the Ti content, the strontium titanate dielectric layer **1110** is a strontium-rich (Sr Rich) layer, the K value will be great and the storage density is high. But the strontium titanate dielectric layer **1110** is easily cracked. Thus, the electric field is not distributed uniformly in the formed capacitor, which results in an electric leakage of the capacitor **11**. Moreover, since Sr is a heavy metal which is easy to form an oxygen vacancy, thus the electric leakage is further increased. When the strontium titanate dielectric layer **1110** is integrally designed with high Ti content, i.e., the Ti content in the overall design of the strontium titanate dielectric layer **1110** is greater than the Sr content, the strontium titanate dielectric layer **1110** is a titanium-rich (Ti Rich) layer, the strontium titanate dielectric layer is not easily cracked, and has a low leakage effect. But the K value will be less, which is harmful for size reduction of the capacitor, thereby being further harmful to improving the storage density of the DRAM chip.

[0051] Therefore, in the embodiment of the present application, the structure of the strontium titanate dielectric layer **1110** is designed gradiently. Specifically, in the direction from the center of the strontium titanate dielectric layer **1110** to the two opposite sides of the strontium titanate dielectric layer **1110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually (i.e., the ratios of the Sr contents in the strontium titanate dielectric layer **1110** decrease gradually and the ratios of the Ti content increase gradually); in other words, in the direction from the two opposite sides of the strontium titanate dielectric layer **1110** to the center of the strontium titanate dielectric layer **1110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** increase gradually (i.e., the ratios of the Sr contents in the strontium titanate dielectric layer **1110** increase gradually and the ratios of the Ti content decrease gradually).

[0052] One side of the two opposite sides of the abovementioned strontium titanate dielectric layer **1110** is close to the first electrode layer **110** and the other side thereof is close to the second electrode layer **112**. In the direction from the center of the strontium titanate dielectric layer **1110** to the first electrode layer **110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually, and in the direction from the center of the strontium titanate dielectric layer **1110** to the second electrode layer **112**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually as well.

[0053] In the solution of the embodiment, compared with the technical solution that the strontium titanate dielectric layer **1110** is integrally the titanium rich layer or the strontium rich layer, the ratios of the Sr contents in the strontium titanate dielectric layer **1110** decrease gradually from the center of the strontium titanate dielectric layer **1110** to the direction of the two opposite sides of the strontium titanate dielectric layer **1110** and the ratios of the Ti contents increase gradually from the center of the strontium titanate dielectric layer **1110** to the direction of the two opposite sides of the strontium titanate dielectric layer **1110**, so that while the high Sr content in the overall strontium titanate dielectric layer **1110** is guaranteed to have a relatively high K value, the stress of the strontium titanate dielectric layer **1110** can be further favorably released or relaxed as it is easy to crystallize due to the high Sr content, i.e., the stress is gradually released from the center of the strontium titanate dielectric layer **1110** to both sides of the strontium titanate dielectric layer; by reducing the internal stress of the strontium titanate dielectric layer **1110**, cracks of the strontium titanate dielectric layer **1110** can be reduced, the surface uniformity of the strontium titanate dielectric layer **1110** is improved, and therefore, it has the low leakage performance.

[0054] That is to say, in the solution, the K value of the capacitor **11** is improved, i.e., the electric leakage is reduced while the storage density of the capacitor **11** is effectively improved, which is beneficial to reducing the critical size in the 1T1C structure and improving the storage density of the three-dimensional DRAM.

[0055] Exemplarily, the ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer **1110** is greater than 50%, i.e., the Sr content at the center of the strontium titanate dielectric layer **1110** is greater than the Ti content, the center of the strontium titanate dielectric layer **1110** is in the Sr rich state to guarantee that the center of the strontium titanate dielectric layer **1110** has the relatively high K value, so as to guarantee that the whole strontium titanate dielectric layer **1110** has a good storage density.

[0056] It shall be noted that the ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer **1110** may be related to its thickness to guarantee the overall performance of the strontium titanate dielectric layer **1110**.

[0057] The thickness of the strontium titanate dielectric layer **1110** may range from 5 nm to 7 nm, for example, 5 nm, 5.5 nm, 6 nm, 7 nm and the like, and the ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer **1110** ranges from 60% to 65%, for example, 60%, 61%, 62%, 63%, 64%, 65% and the like.

[0058] For example, when the thickness of the strontium titanate dielectric layer **1110** is about 6 nm, the ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer **1110** may substantially reach 62%, so that the capacitor may realize the performance of the high K value (>60) and low electric leakage (<10.sup.−5).

[0059] In an embodiment of the present application, as shown in FIG. **7**, the strontium titanate dielectric layer **1110** is of a single-layer structure as a whole, that is to say, when the strontium titanate dielectric layer **1110** is formed at a side of the first electrode layer **110** away from the substrate **10**, it may be formed by using a film forming process, for example, a strontium titanate material may be deposited at the side of the first electrode layer **110** away from the substrate **10** through an Atomic layer deposition (ALD) process to form the strontium titanate dielectric layer **1110**.

[0060] In the atomic layer deposition process, the ratios of Sr contents in the strontium titanate dielectric layer **1110** increase gradually and decrease gradually successively by adjusting the flow of an Sr precursor and/or a Ti precursor, so that the ratios of Sr/(Sr+Ti) in the strontium titanate material increase gradually and decrease gradually successively to form the abovementioned strontium titanate dielectric layer **1110**, i.e., in the direction from the two opposite sides of the strontium titanate dielectric layer **1110** to the center of the strontium titanate dielectric layer **1110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** increase gradually.

[0061] Further, as shown in FIG. **7**, in the strontium titanate dielectric layer **1110**, a surface close to the first electrode layer **110** is defined as a first boundary surface a**1**, a surface close to the second electrode layer **112** is defined as a second boundary surface a**2**, and the content of Sr at the first boundary surface a**1** and the second boundary surface a**2** may be 0; in other words, when the strontium titanate dielectric layer **1110** is manufactured, the initial strontium titanate material deposited in the primary atomic layer deposition process does not contain Sr or the Sr content is very small and is approximately 0; then the Sr ratio in the strontium titanate material is increased gradually; when it reaches a set quantity, the Sr ratio in the strontium titanate material is decreased gradually, so that the Sr content in the strontium titanate material when the primary atomic layer deposition process is finished is 0 or is approximately 0.

[0062] It shall be appreciated that when the content of Sr at the first boundary surface a**1** and the second boundary surface a**2** in the strontium titanate dielectric layer **1110** is 0, the first boundary surface a**1** and the second boundary surface a**2** may be construed as titanium dioxide (TiO.sub.2) surfaces, and TiO.sub.2 used at the boundary surfaces may effectively reduce the oxygen vacancies in the strontium titanate dielectric layer **1110**, so that electric leakage is reduced.

[0063] For example, when the strontium titanate dielectric layer **1110** of the single-layer structure is manufactured, in addition to containing Sr, the strontium titanate may further be doped with aluminum (Al) when the primary atomic layer deposition process is started and finished, that is to say, the first boundary surface a**1** and the second boundary surface a**2** may be aluminum titanate (ATO) boundary surfaces to improve the conduction band offset (CBO), so as to further reduce electric leakage.

[0064] It shall be appreciated that when a semiconductor and a semiconductor or an insulator and a semiconductor are in contact to form an interfacial structure, due to different energy gap widths, discontinuous offsets will be formed at the bottom of a conduction band and at the top of a valence band of materials on both sides, i.e., band offset. The band offset at the bottom of the conduction band is called a conduction band offset (CBO) and the band offset at the top of the valence band is called a valence band offset (VBO).

[0065] In an optional embodiment, in the direction from the center (the center passes through the dotted line in FIG. **7**) of the strontium titanate dielectric layer **1110** to the two opposite sides of the strontium titanate dielectric layer **1110** (the direction shown by a black bold arrow in FIG. **7**), the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually according to a linear relation, and in other words, in the direction from the center of the strontium titanate dielectric layer **1110** to the two opposite sides of the strontium titanate dielectric layer **1110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** may decrease at a constant speed to better balance the performance of the strontium titanate dielectric layer **1110** with high K value and low electric leakage.

[0066] Exemplarily, as shown in FIG. **8**, the abovementioned linear relation may specifically be:

[00001] $\frac{Sr}{Sr+Ti} = C - \frac{2C \times D}{H}$, Equation(1)

[0067] C is a ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer **1110**, D is a

distance deviated from the center of the strontium titanate dielectric layer **1110**, H is a thickness of the strontium titanate dielectric layer **1110**, and 0≤D≤H/2.
[0068] For example, when C is 0.62 and His 6 nm, the linear relation may be:
[00002] $\frac{Sr}{Sr+Ti} = 0.62 - \frac{0.62 \times D}{3}$, [0069] when the distance deviated from the center of the strontium titanate dielectric layer **1110** in the direction close to the first electrode layer **110** is 3 nm, the ratio of Sr/(Sr+Ti) is 0, which may be construed as the abovementioned first boundary surface a**1**; and when the distance deviated from the center of the strontium titanate dielectric layer **1110** in the direction close to the second electrode layer **112** is 3 nm, the ratio of Sr/(Sr+Ti) is 0, which may be construed as the abovementioned second boundary surface a**2**.
[0070] In another optional embodiment, as shown in FIG. **9**, in the direction from the center of the strontium titanate dielectric layer **1110** to the two opposite sides of the strontium titanate dielectric layer **1110**, the decreasing rates of the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** increase gradually, and in other words, in the direction from the two opposite sides of the strontium titanate dielectric layer **1110** to the center of the strontium titanate dielectric layer **1110**, the increasing rates of the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually, that is to say, when the whole strontium titanate dielectric layer **1110** is manufactured, the K value of the strontium titanate dielectric layer **1110** may further be improved while guaranteeing the strontium titanate dielectric layer **1110** has the performance of low electric leakage by rapidly increasing the Sr content first, then slowly increasing the Sr content to the maximum value, and then slowly decreasing the Sr content and rapidly decreasing the Sr content.
[0071] Exemplarily, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** meet the following relational expression:
[00003] $\frac{Sr}{Sr+Ti} = C \times \cos(\frac{\times D}{H})$, Equation(2) [0072] C is the ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer **1110**, D is a distance deviated from the center of the strontium titanate dielectric layer **1110**, H is a thickness of the strontium titanate dielectric layer **1110**, and 0≤D≤H/2.
[0073] For example, when C is 0.62 and His 6 nm, the relational expression met by the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** may be:
[00004] $\frac{Sr}{Sr+Ti} = 0.62 \times \cos(\frac{D}{6})$, [0074] when the distance deviated from the center of the strontium titanate dielectric layer **1110** in the direction close to the first electrode layer **110** is 3 nm, the ratio of Sr/(Sr+Ti) is 0, which may be construed as the abovementioned first boundary surface a**1**; and when the distance deviated from the center of the strontium titanate dielectric layer **1110** in the direction close to the second electrode layer **112** is 3 nm, the ratio of Sr/(Sr+Ti) is 0, which may be construed as the abovementioned second boundary surface a**2**.
[0075] In yet another optional embodiment, as shown in FIG. **10**, in the direction from the center of the strontium titanate dielectric layer **1110** to the two opposite sides of the strontium titanate dielectric layer **1110**, the decreasing rates of the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually, and in other words, in the direction from the two opposite sides of the strontium titanate dielectric layer **1110** to the center of the strontium titanate dielectric layer **1110**, the increasing rates of the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** increase gradually, that is to say, when the whole strontium titanate dielectric layer **1110** is manufactured, the surface uniformity of the two opposite sides of the strontium titanate dielectric layer **1110** may further be improved while guaranteeing the strontium titanate dielectric layer **1110** has the K value by slowly increasing the Sr content first, then rapidly increasing the Sr content to the maximum value, and then rapidly decreasing the Sr content and slowly decreasing the Sr content, so that the strontium titanate dielectric layer has the performance of low electric leakage.
[0076] It shall be appreciated that the X axis in FIGS. **8-10** is used to represent the distance deviated from the strontium titanate dielectric layer **1110**, with the unit of nm (nanometer), the Y axis is used to represent the ratio of Sr/(Sr+Ti), the line on the left side of the Y axis is used to

represent the ratio change of Sr/(Sr+Ti) between the center of the strontium titanate dielectric layer **1110** and the first electrode layer **110**, and the line on the right side of the Y axis is used to represent the ratio change of Sr/(Sr+Ti) between the center of the strontium titanate dielectric layer **1110** and the second electrode layer **112**.
[0077] Besides, in order to further improve the performance of the capacitor, as shown in FIG. **9**, the first electrode layer **110** and the second electrode layer **112** in the embodiment each may include a ruthenium element film layer **11***a*, the ruthenium element film layer **11***a* is a metal ruthenium (Ru) film layer or a ruthenium oxide (RuO.sub.2) film layer with a high work function, where the work function is also called a work function and escape work, which is defined in solid physics as follows: least energy needed to just move an electron from the inside of a solid to the surface of the object.
[0078] For example, the dielectric layer **111** of the capacitor may only include the abovementioned strontium titanate dielectric layer **1110**, and the first boundary surface a**1** of the strontium titanate dielectric layer **1110** may be in contact with the ruthenium element film layer **11***a* of the first electrode layer **110**, the second boundary surface a**2** of the strontium titanate dielectric layer **1110** may be in contact with the ruthenium element film layer **11***a* of the second electrode layer **112**, and when the first boundary surface a**1** and the second boundary surface a**2** of the strontium titanate dielectric layer **1110** are defined as the TiO.sub.2 surfaces, the ruthenium element film layer **11***a* may promote formation of a rutile structure by TiO.sub.2 at the boundary surfaces, thereby improving the K value while reducing the electric leakage. Therefore, the K value of the strontium titanate dielectric layer **1110** decreases first and then increases from the middle to both sides, the dielectric constant is mainly improved in the middle and at boundaries, and the area between the middle and the boundaries mainly plays a role in reducing the electric leakage.
[0079] As shown in FIG. **9**, the first electrode layer **110** and the second electrode layer **112** each further include a titanium nitride film layer **11***b*, and the titanium nitride film layer **11***b* is formed at a side of the ruthenium element film layer **11***a* away from the strontium titanate dielectric layer **1110**, so that the thickness of the ruthenium element film layer **11***a* may be properly reduced by arranging the titanium nitride film layer **11***b* while guaranteeing that the overall thickness and work functions of the first electrode layer **110** and the second electrode layer **112** meet the requirements. Therefore, the cost of the first electrode layer **110** and the second electrode layer **112** is properly reduced while TiO.sub.2 at the boundary surfaces may be promoted to form the rutile structure to reduce the electric leakage and improve the K value.
Embodiment II
[0080] Compared with Embodiment I, the main difference between the embodiment in the present disclosure and Embodiment I lies in that the strontium titanate dielectric layer **1110** in the capacitor in Embodiment I is designed in the single-layer structure, and the strontium titanate dielectric layer **1110** in the capacitor in Embodiment II may be of a multilayer stack structure.
[0081] The structure of the capacitor in Embodiment of the present application will be descried in detail below in conjunction with the accompanying drawings.
[0082] In the embodiment, as shown in FIG. **12**, the strontium titanate dielectric layer **1110** may be of the multilayer stack structure and the strontium titanate dielectric layer **1110** may include N strontium titanate film layers **11101** stacked in sequence, N being a positive integer greater than or equal to 3; it shall be appreciated that a film layer is formed by each forming process, that is to say, the strontium titanate dielectric layer **1110** is formed by using multiple forming processes. For example, the step of forming the strontium titanate dielectric layer **1110** at the side of the first electrode layer **110** away from the substrate **10** may specifically include: N strontium titanate film layers **11101** stacked in sequence are formed at the side of the first electrode layer **110** away from the substrate **10** through the N atomic layer deposition processes to form the stacked strontium titanate dielectric layer **1110**.
[0083] It shall be noted that the strontium titanate film layer **11101** in the embodiment is a film

layer containing not only Sr, but also Ti, that is to say, the ratios of Sr/(Sr+Ti) in each strontium titanate film layer **11101** are greater than 0.

[0084] In the direction from the center of the strontium titanate dielectric layer **1110** to the two opposite sides of the strontium titanate dielectric layer **1110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease layer by layer. As shown in FIG. **12**, taking N=3 as an example for description, the strontium titanate dielectric layer **1110** may include the first strontium titanate film layers **11101**, the second strontium titanate film layers **11101** and the third strontium titanate film layers **11101** stacked in sequence, where the ratios of Sr/(Sr+Ti) in the second strontium titanate film layers **11101** are the greatest, and the ratios of Sr/(Sr+Ti) in the first strontium titanate film layers **11101** and the third strontium titanate film layers **11101** are less than the ratios of Sr/(Sr+Ti) in the second strontium titanate film layers **11101**. It shall be noted that the strontium titanate dielectric layer **1110** is not limited to include the three strontium titanate film layers **11101** shown in FIG. **12**, and further includes 4, 5 layers and the like, which depends on circumstances.

[0085] In the embodiment, the strontium titanate dielectric layer **1110** is designed as the multilayer stack structure, so that the manufacturing mode is simple while the gradient design of the strontium titanate dielectric layers **1110** is realized to guarantee that the strontium titanate dielectric layers **1110** has the characteristics of high K value and low electric leakage. Furthermore, while it is guaranteed that the thickness of the strontium titanate dielectric layers **1110** is unchanged, the thickness of each layer is less through layered manufacturing, so that the film forming uniformity is easily controlled, and thus, the product performance is improved. Moreover, it is beneficial for the product to come into use rapidly. The strontium titanate film layers **11101** may be formed in sequence by using known equipment without researching and developing the equipment again, so that the research and development costs are reduced.

[0086] Besides, in the embodiment, the strontium titanate dielectric layer **1110** is designed being formed by stacking at least three strontium titanate film layers **11101**, so that the gradient design of the area of the strontium titanate dielectric layer **1110** to the first electrode layer **110** and the area of the strontium titanate dielectric layer close to the second electrode layer **112** is realized, guaranteeing the characteristics of high K value and low electric leakage between the strontium titanate dielectric layer **1110** and the first electrode layer **110** and between the strontium titanate dielectric layer and the second electrode layer **112**.

[0087] Exemplarily, N in the embodiment may be an odd number greater than or equal to 3, that is to say, the quantity of the strontium titanate film layers **11101** in the strontium titanate dielectric layer **1110** is an odd number which is greater than or equal to 3, for example, N=3, 5, 7, 9 and the like, which depends on circumstances.

[0088] In the embodiment, the quantity of the strontium titanate film layers **11101** in the strontium titanate dielectric layer **1110** is designed as the odd number, so that it is guaranteed that the quantities of the film layers on the two opposite sides of the strontium titanate film layer **11101** in the middle are same to guarantee the balanced gradient design in the area of the strontium titanate dielectric layer **1110** to the first electrode layer **110** and the area of the strontium titanate dielectric layer close to the second electrode layer **112**, thereby improving the product quality.

[0089] The ratios of Sr/(Sr+Ti) in all parts of the strontium titanate film layers **11101** may be equal, that is to say, in each atomic layer deposition process, the constants of the Sr precursor and the Ti precursor are constantly unchanged, so that the strontium titanate film layers **11101** may be formed in sequence by using the known equipment without researching and developing the equipment again, thereby reducing the research and development costs.

[0090] It shall be appreciated that the ratios of Sr/(Sr+Ti) among the strontium titanate film layers **11101** in the strontium titanate dielectric layer **1110** may be designed according to the equations (1) and (2) mentioned in Embodiment I, where in Embodiment II, the parameter C mentioned in the equations (1) and (2) is the ratio of Sr/(Sr+Ti) in the middlemost strontium titanate film layer **11101** and the parameter D is the distance from the center of another film layer deviated from the center of the middlemost strontium titanate film layer **11101**.

[0091] In an optional embodiment, as shown in FIG. **13**, besides the abovementioned strontium titanate dielectric layer **1110**, the dielectric layer **111** of the capacitor **11** may further include a first titanium oxide dielectric layer **1111** and a second titanium oxide dielectric layer **1112**, the first titanium oxide dielectric layer **1111** being formed between the first electrode layer **110** and the strontium titanate film layer **11101** of the strontium titanate dielectric layer **1110** and the second titanium oxide dielectric layer **1112** being formed between the second electrode layer **112** and the strontium titanate dielectric layer **1110**. By forming the first titanium oxide dielectric layer **1111** and the second titanium oxide dielectric layer **1112** at the two opposite sides of the strontium titanate dielectric layer **1110**, the gradient design of the dielectric layer **111** is further realized, i.e., compared with the middle strontium titanate film layer **11101**, the film layers located at the two outermost sides of the dielectric layer **111** do not contain Sr to be formed as the first titanium oxide dielectric layer **1111** and the second titanium oxide dielectric layer **1112**. Therefore, the uniformity of the surfaces of the two outermost sides of the dielectric layer **111** as a whole in the capacitor **11** may be guaranteed, so that the capacitor has the performance of low electric leakage.

[0092] Besides, when the first electrode layer **110** and the second electrode layer **112** in the embodiment each include the ruthenium element film layer **11***a* mentioned in the above embodiment, the first titanium oxide dielectric layer **1111** may be in contact with the ruthenium element film layer **11***a* of the first electrode layer **110**, and the second titanium oxide dielectric layer **1112** may be in contact with the ruthenium element film layer **11***a* of the second electrode layer **112**. The ruthenium element film layer **11***a* may promote the first titanium oxide dielectric layer **1111** and the second titanium oxide dielectric layer **1112** to be in contact with the ruthenium element film layer to form the rutile structure, so that the K value is improved while electric leakage is reduced. Therefore, the K value in the whole dielectric layer **111** decreases first and then increases from the middle to both sides in the first electrode layer **110** and the second electrode layer **112**. The dielectric constant is mainly improved in the middle and at boundaries, and the area between the middle and the boundaries mainly plays a role in reducing the electric leakage.

[0093] In another optional embodiment, as shown in FIG. **14**, besides the abovementioned strontium titanate dielectric layer **1110**, the dielectric layer **111** of the capacitor **11** may further include a first aluminum titanate dielectric layer **1113** and a second aluminum titanate dielectric layer **1114**, the first aluminum titanate dielectric layer **1113** being formed between the first electrode layer **110** and the strontium titanate dielectric layer **1110** and the second aluminum titanate dielectric layer **1114** being formed between the second electrode layer **112** and the strontium titanate dielectric layer **1110**.

[0094] For example, the first aluminum titanate dielectric layer **1113** may be in contact with the ruthenium element film layer **11***a* of the first electrode layer **110** and the second aluminum titanate dielectric layer **1114** may be in contact with the ruthenium element film layer **11***a* of the second electrode layer **112**. Therefore, in such a design, the conduction band offset (CBO) may be improved while improving the K value, so that the electric leakage is further reduced.

[0095] It shall be noted that except for the above difference with Embodiment I, other designs in Embodiment II may refer to the content in Embodiment I, which are not repeatedly described here.

Embodiment III

[0096] The embodiment of the present application further provides a manufacturing method of a memory, which may manufacture the memory provided in the above embodiments, and detailed description about the memory may refer to the above embodiments, which is not repeatedly described here.

[0097] In conjunction with FIGS. **1** to **25**, the manufacturing method of the memory may at least include step S**100**, step S**200**, step S**300** and step S**400**.

[0098] In step S**100**, a substrate **10** is provided. For example, a semiconductor base **100** may be provided first, and then an insulation isolation layer **101** is formed on the semiconductor base **100**, as shown in FIGS. **2-5**. The semiconductor base **100** may be a silicon substrate, the insulation isolation layer **101** may be SiO.sub.2, and the deposition thickness of the insulation isolation layer **101** may be 2000 Å (angstrom).

[0099] In step S**200**, a first electrode **110** is formed on the substrate **10**. For example, the first electrode layer **110** may be formed at a side of the insulation isolation layer **101** away from the semiconductor base **100**, where the first electrode layer **110** may include a TiN layer and a RuO.sub.2 layer formed at the side of the insulation isolation layer **101** away from the semiconductor base **100** in sequence.

[0100] In step S**300**, a dielectric layer **111** is formed at a side of the first electrode layer **110** away from the substrate **10**, and the dielectric layer **111** at least includes a strontium titanate dielectric layer **1110**. In a direction from a center of the strontium titanate dielectric layer **1110** to two opposite sides of the strontium titanate dielectric layer **1110**, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer **1110** decrease gradually. For example, the strontium titanate dielectric layer **1110** may be formed at a side of the RuO.sub.2 layer away from the semiconductor base **100**.

[0101] In an optional embodiment, step S**300** specifically includes: depositing a strontium titanate material at a side of the first electrode layer **110** away from the substrate **10** through a primary atomic layer deposition process to form the strontium titanate dielectric layer **1110**, as shown in FIG. **7**; in the primary atomic layer deposition process, the ratios of Sr/(Sr+Ti) in the strontium titanate material increasing gradually and decreasing gradually successively according to a set rule by adjusting the flow of an Sr precursor and/or a Ti precursor.

[0102] In another optional embodiment, step S**300** specifically includes: forming N strontium titanate film layers **11101** stacked in sequence at the side of the first electrode layer **110** away from the substrate **10** through N atomic layer deposition processes to form the strontium titanate dielectric layer **1110**, N being a positive integer greater than or equal to 3, as shown in FIGS. **12-14**; in each atomic layer deposition process, the flows of the Sr precursor and the Ti precursor may be constantly unchanged substantially, i.e., within an error range.

[0103] In step S**400**, a second electrode layer **112** is formed at a side of the strontium titanate dielectric layer **1110** away from the first electrode layer **110** to form a capacitor. The second electrode layer **112** may include a RuO.sub.2 layer and a TiN layer formed at a side of the strontium titanate dielectric layer **1110** away from the first electrode layer **110** in sequence.

[0104] In an optional embodiment, before step S**300** and after step S**200**, the manufacturing method further includes: forming a first titanium oxide dielectric layer **1111** at a side of the first electrode layer **110** away from the substrate **10**; and after step S**300** and before step S**400**, the manufacturing method further includes: forming a second titanium oxide dielectric layer **1112** at a side of the strontium titanate dielectric layer **1110** away from the first electrode layer **110**, as shown in FIG. **13**.

[0105] In another optional embodiment, before step S**300** and after step S**200**, the manufacturing method further includes: forming a first aluminum titanate dielectric layer **1113** at a side of the first electrode layer **110** away from the substrate **10**; and after step S**300** and before step S**400**, the manufacturing method further includes: forming a second aluminum titanate dielectric layer **1114** at a side of the strontium titanate dielectric layer **1110** away from the first electrode layer **110**, as shown in FIG. **14**.

[0106] The step of forming a first aluminum titanate dielectric layer **1113** at a side of the first electrode layer **110** away from the substrate **10** may specifically include: forming the first titanium oxide dielectric layer **1111** at the side of the first electrode layer **110** away from the substrate **10**, and doping an aluminum element into the first titanium oxide dielectric layer **1111** to form the first aluminum titanate dielectric layer **1113**; and the step of forming a second aluminum titanate dielectric layer **1114** at a side of the strontium titanate dielectric layer **1110** away from the first electrode layer **110** includes: forming the second titanium oxide dielectric layer **1112** at the side of the strontium titanate dielectric layer **1110** away from the first electrode layer **110**, and then doping the aluminum element into the second titanium oxide dielectric layer **1112** to form the second aluminum titanate dielectric layer **1114**.

[0107] It shall be appreciated that the aluminum element may be doped at the boundary surface of the first titanium oxide dielectric layer **1111** toward the first electrode layer **110** to form the first aluminum titanate dielectric layer **1113** and certain space may also be expanded inward. Similarly, the aluminum element may also be doped at the boundary surface of the second titanium oxide dielectric layer **1112** toward the second electrode layer **112** to form the second aluminum titanate dielectric layer **1114**, and certain space may also be expanded inward.

[0108] Before step S**300** and after step S**200**, the manufacturing method further includes: forming an interlevel dielectric material film **15***a* at a side of the first electrode layer **110** away from the substrate **10**, as shown in FIG. **15**; and forming a through hole **150** through which part of the first electrode layer **110** is exposed in the interlevel dielectric material film **15***a* through a patterning treatment process to form an interlevel dielectric layer **15**.

[0109] For example, a photoresist layer may first be formed at a side of the interlevel material film **15***a* away from the first electrode layer **110**, and then the photoresist layer is subjected to exposure and development to form a first photoresist mask layer **19**, as shown in FIG. **16**; then, the interlevel material film **15***a* is etched with the first photoresist mask layer **19** to form the through hole **150**, and finally, the first photoresist mask layer **19** is removed, as shown in FIG. **17**.

[0110] It shall be appreciated that at least part of the strontium titanate dielectric layer **1110** is located in the through hole **150** and is in contact with the first electrode layer **110**.

[0111] The step of forming the strontium titanate dielectric layer **1110** and the second electrode layer **112** at the side of the first electrode layer **110** away from the substrate **10** may specifically include:

[0112] After forming the interlevel dielectric layer **15**, forming a dielectric material film (the strontium titanate dielectric material film) **111***a* and a second electrode material film **112***a* located in the through hole **150** and covering the interlevel dielectric layer **15** in sequence, as shown in FIG. **18**; and then removing the part of the dielectric material film (the strontium titanate dielectric material film) **111***a* and the second electrode material film **112***a* covering the interlevel dielectric layer **15** to form the strontium titanate dielectric layer **1110** and the second electrode layer **112**, as shown in FIG. **20**.

[0113] For example, the photoresist layer may first be formed on the second electrode material film **112** and is subjected to exposure and development to form a second photoresist mask layer **20**. Then, the dielectric material film (the strontium titanate dielectric material film) **111***a* and the second electrode material film **112***a* are etched with the second photoresist mask layer **20** to remove the part of the dielectric material film (the strontium titanate dielectric material film) **111***a* and the second electrode material film **112***a* covering the interlevel dielectric layer **15** to form the strontium titanate dielectric layer **1110** and the second electrode layer **112**, as shown in FIG. **19**. Finally, the second photoresist mask layer **20** is removed, as shown in FIG. **20**.

[0114] After step S**400**, the manufacturing method further includes step S**500**, step S**600**, step S**700** and step S**800**.

[0115] In step S**500**, an insulation dielectric layer **12** and an etch stop layer **13** are formed in sequence at the sides of the interlevel dielectric layer **15** and the second electrode layer **112** away from the substrate **10**, as shown in FIG. **21**. For example, a material of the insulation dielectric layer **12** may be SiO.sub.2, a material of the etch stop layer **13** may be SiN, which is not limited thereto. The materials of the insulation dielectric layer **12** and the etch stop layer **13** may also be selected according to the actual condition.

[0116] In step S**600**, a first via hole **21** and a second via hole **22** are formed at an interval, as shown in FIG. **23**. An orthographic projection of the first via hole **21** in the substrate **10** falls within an orthographic projection of an edge area of the first electrode layer **110** on the substrate **10**, and does not overlap with orthographic projections of the second electrode layer **112** and the strontium titanate dielectric layer **1110** on the substrate **10**, the first via hole **21** penetrates through the insulation dielectric layer **12**, the etch stop layer **13** and the interlevel dielectric layer **15** in sequence and part of the first electrode layer **110** is exposed, an orthographic projection of the second via hole **22** in the substrate **10** falls within an orthographic projection of the through hole **150** in the substrate **10**, the second via hole **22** penetrates through the insulation dielectric layer **12** and the etch stop layer **13** in sequence, and part of the second electrode layer **112** is exposed.

[0117] For example, a photoresist layer may first be formed at a side of the etch stop layer **13** away from the insulation dielectric layer **12**, and then the photoresist layer is subjected to exposure and development to form a third photoresist mask layer **23**, as shown in FIG. **22**; then, the photoresist layer is etched with the third photoresist mask layer **23** to form the first via hole **21** and the second via hole **22**, and finally, the third photoresist mask layer **23** is removed, as shown in FIG. **23**.

[0118] In step S**700**, a measurement electrode material film **14** is formed. The measurement electrode material film **14** cover a surface of the etch stop layer **13** away from the insulation dielectric layer **12** and fills the first via hole **21** and the second via hole **22**, as shown in FIG. **24**. For example, a material of the measurement electrode material film **14** may be TiN, but is not limited thereto. It may also be other conducting materials.

[0119] In step S**800**, patterning treatment on the measurement electrode material film **14** to form a first measurement electrode **14***a* and a second measurement electrode **14***b*, where the first measurement electrode **14***a* and the second measurement electrode **14***b* each include a via hole conduction portion **140** and a measurement conduction portion **141**, the measurement conduction portion **141** of the first measurement electrode **14***a* and the measurement conduction portion of the second measurement electrode **14***b* are arranged at intervals and are both formed at the side of the etch stop layer **13** away from the substrate **10**, the measurement conduction portion **141** of the first measurement electrode **14***a* is a measurement electrode material film **14** filling the first via hole **21**, and the measurement conduction portion **141** of the second measurement electrode **14***b* is a measurement electrode material film **14** filling the second via hole **22**, as shown in FIG. **5**.

[0120] For example, a photoresist layer may first be formed at a side of the measurement electrode material film **14** away from the etch stop layer **13**, and then the photoresist layer is subjected to exposure and development to form a fourth photoresist mask layer **24**; then, the measurement electrode material film **14** on the surface of the etch stop layer **13** is etched with the fourth photoresist mask layer **24**, as shown in FIG. **25**, to form the measurement conduction portion **141** of the first measurement electrode **14***a* and the second measurement electrode **14***b*, and finally, the fourth photoresist mask layer **24** is removed, as shown in FIG. **25**.

[0121] After the first measurement electrode **14***a* and the second measurement electrode **14***b* are formed, an external measuring device is connected to the first measurement electrode **14***a* and the second measurement electrode **14***b* to measure the performance of the capacitor.

[0122] It shall be appreciated that after the performance of the capacitor is tested with the first measurement electrode **14***a* and the second measurement electrode **14***b*, the first measurement electrode **14***a* and the second measurement electrode **14***b* may either be cut during die cutting or be reserved on the substrate **10**, which depends on circumstances.

[0123] In addition, the terms “first”, “second”, “third” and the like are only used for a description purpose rather than being construed to indicate or imply relative importance or implicitly indicate the quantity of indicated technical features. Thus, features defining “first”, “second” and “third” may expressively or implicitly include one or more features. In the description of the present application, “a plurality of” means two or more, unless expressly specified otherwise.

[0124] In the description of the specification, the description with reference to the terms “some embodiments”, “exemplarily”, and the like means that specific features, structures, materials, or features described in connection with the embodiments or examples are included in at least one embodiment or example of the present application. In the description, schematic expressions of the terms do not have to mean same embodiments or exemplary embodiments. Furthermore, specific features, structures, materials or characteristics described may be combined in any one or more embodiments or exemplary embodiments in proper manners. In addition, under a condition without mutual contradiction, those skilled in the art may integrate or combine different embodiments or exemplary embodiments with different embodiments or exemplary embodiments described in the description.

[0125] Although the embodiments of the present application have been shown and described above, it may be appreciated that the embodiments are exemplary and cannot be construed as a limitation to the present application. Those of ordinary skill in the art can make changes, modification, replacement and transformation on the embodiments within the scope of the present application. Changes and modifications made according to claims and description of the present application shall fall into the scope of the present application.

Claims

1. A capacitor, comprising: a first electrode layer; a second electrode layer; and a strontium titanate dielectric layer, formed between the first electrode layer and the second electrode layer; wherein in a direction from a center of the strontium titanate dielectric layer to two opposite sides of the strontium titanate dielectric layer, ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually; and one side of the two opposite sides in the strontium titanate dielectric layer is close to the first electrode layer and the other side of the two opposite sides in the strontium titanate dielectric layer is close to the second electrode layer.

2. The capacitor according to claim 1, wherein: the strontium titanate dielectric layer is of a multilayer stacked structure and comprises N strontium titanate film layers stacked in sequence, N is a positive integer greater than or equal to 3; and in the direction from the center of the strontium titanate dielectric layer to the two opposite sides of the strontium titanate dielectric layer, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease layer by layer.

3. The capacitor according to claim 2, wherein the ratios of Sr/(Sr+Ti) at everywhere of the strontium titanate dielectric layer are equal.

4. The capacitor according to claim 1, wherein in the direction from the center of the strontium titanate dielectric layer to the two opposite sides of the strontium titanate dielectric layer, the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually according to a linear relation.

5. The capacitor according to claim 4, wherein the linear relation is specifically as follows: $\frac{Sr}{Sr+Ti} = C - \frac{2C \times D}{H}$, wherein C is a ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer, D is a distance deviated from the center of the strontium titanate dielectric layer in the strontium titanate dielectric layer, H is a thickness of the strontium titanate dielectric layer, and 0≤D≤H/2.

6. The capacitor according to claim 1, wherein in the direction from the center of the strontium titanate dielectric layer to the two opposite sides of the strontium titanate dielectric layer, decreasing rates of the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer increase gradually.

7. The capacitor according to claim 6, wherein the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer meet the following relational expression: $\frac{Sr}{Sr+Ti} = C \times \cos(\frac{\times D}{H})$, wherein C is a ratio of Sr/(Sr+Ti) at the center of the strontium titanate dielectric layer, D is a distance deviated from the center of the strontium titanate dielectric layer, H is a thickness of the strontium titanate dielectric layer, and 0≤D≤H/2.

8. The capacitor according to claim 1, wherein in the direction from the center of the strontium titanate dielectric layer to the two opposite sides of the strontium titanate dielectric layer,

decreasing rates of the ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually.
9. A memory, comprising a substrate and the capacitor of claim 1, wherein the capacitor is formed on the substrate.
10. The memory according to claim 9, wherein an orthographic projection of the second electrode layer on the first electrode layer is overlapped with an orthographic projection of the strontium titanate dielectric layer on the first electrode layer, orthographic projections of the second electrode layer and the strontium titanate dielectric layer on the substrate fall within a middle area of an orthographic projection of the first electrode layer on the substrate, and the memory further comprises: an insulation dielectric layer, covering an edge area of the first electrode layer and the second electrode layer; an etch stop layer, formed at a side of the insulation dielectric layer away from the substrate; and a first measurement electrode and a second measurement electrode arranged at intervals, the first measurement electrode and the second measurement electrode each comprising a via hole conduction portion and a measurement conduction portion connected with each other, wherein: the measurement conduction portion of the first measurement electrode and the measurement conduction portion of the second measurement electrode are formed at a side of the etch stop layer away from the substrate; an orthographic projection of the via hole conduction portion of the first measurement electrode on the substrate falls within an orthographic projection of the edge area of the first electrode layer on the substrate, and the via hole conduction portion of the first measurement electrode penetrates through the etch stop layer and the insulation dielectric layer in sequence and is in contact with the edge area of the first electrode layer; and an orthographic projection of the via hole conduction portion of the second measurement electrode on the substrate falls within the orthographic projection of the second electrode layer on the substrate, and the via hole conduction portion of the second measurement electrode penetrates through the etch stop layer and the insulation dielectric layer in sequence and is in contact with the second electrode layer.
11. The memory according to claim 10, further comprising an interlevel dielectric layer, wherein the interlevel dielectric layer is formed between the first electrode layer and the insulation dielectric layer, and the interlevel dielectric layer has a through hole to expose part of the first electrode layer; the strontium titanate dielectric layer has a middle portion and an edge portion arranged around the middle portion, the middle portion is formed in the through hole and in contact with the first electrode layer, and the edge portion is in overlap joint to a surface of the interlevel dielectric layer away from the first electrode layer; the via hole conduction portion of the first measurement electrode further penetrates through the interlevel dielectric layer to be in contact with the edge area of the first electrode layer while penetrating through the etch stop layer and the insulation dielectric layer; and the orthographic projection of the via hole conduction portion of the second measurement electrode on the substrate falls within the orthographic projection of the through hole in the substrate.
12. A manufacturing method of a memory, comprising: providing a substrate; forming a first electrode layer on the substrate; forming a strontium titanate dielectric layer at a side of the first electrode layer away from the substrate, wherein in a direction from a center of the strontium titanate dielectric layer to two opposite sides of the strontium titanate dielectric layer, ratios of Sr/(Sr+Ti) in the strontium titanate dielectric layer decrease gradually, one side of the two opposite sides in the strontium titanate dielectric layer is close to the first electrode layer and the other side of the two opposite sides in the strontium titanate dielectric layer is away from the first electrode layer; and forming a second electrode layer at a side of the strontium titanate dielectric layer away from the first electrode layer to form a capacitor.
13. The manufacturing method according to claim 12, wherein the step of forming the strontium titanate dielectric layer at the side of the first electrode layer away from the substrate comprises: depositing a strontium titanate material at the side of the first electrode layer away from the

substrate through a primary atomic layer deposition process to form the strontium titanate dielectric layer; wherein in the primary atomic layer deposition process, the ratios of Sr/(Sr+Ti) in the strontium titanate material increase gradually and decrease gradually successively according to a set rule by adjusting a flow of an Sr precursor and/or a Ti precursor.

14. The manufacturing method according to claim 12, wherein the step of forming the strontium titanate dielectric layer at the side of the first electrode layer away from the substrate comprises: forming N strontium titanate film layers stacked in sequence at the side of the first electrode layer away from the substrate through N atomic layer deposition processes to form the strontium titanate dielectric layer, wherein N is a positive integer greater than or equal to 3.

15. The manufacturing method according to claim 12, wherein: prior to the step of forming the strontium titanate dielectric layer at the side of the first electrode layer away from the substrate, the manufacturing method further comprises: forming a first titanium oxide dielectric layer at the side of the first electrode layer away from the substrate; and prior to the step of forming the second electrode layer at the side of the strontium titanate dielectric layer away from the first electrode layer, the manufacturing method further comprises: forming a second titanium oxide dielectric layer at the side of the first electrode layer away from the strontium titanate dielectric layer.

16. The manufacturing method according to claim 12, wherein: prior to the step of forming the strontium titanate dielectric layer at the side of the first electrode layer away from the substrate, the method further comprises: forming a first aluminum titanate dielectric layer at the side of the first electrode layer away from the substrate; and prior to the step of forming the second electrode layer at the side of the strontium titanate dielectric layer away from the first electrode layer, the method further comprises: forming a second aluminum titanate dielectric layer at the side of the strontium titanate dielectric layer away from the first electrode layer.

17. The manufacturing method according to claim 16, wherein: the step of forming the first aluminum titanate dielectric layer at the side of the first electrode layer away from the substrate comprises: forming a first titanium oxide dielectric layer at the side of the first electrode layer away from the substrate and then doping an aluminum element into the first titanium oxide dielectric layer to form the first aluminum titanate dielectric layer; and the step of forming the second aluminum titanate dielectric layer at the side of the strontium titanate dielectric layer away from the first electrode layer comprises: forming a second titanium oxide dielectric layer at the side of the strontium titanate dielectric layer away from the first electrode layer and then doping an aluminum element into the second titanium oxide dielectric layer to form the second aluminum titanate dielectric layer.

18. The manufacturing method according to claim 12, wherein prior to the step of forming the strontium titanate dielectric layer at the side of the first electrode layer away from the substrate, the method further comprises: forming an interlevel dielectric material film at the side of the first electrode layer away from the substrate; and forming a through hole through which a middle area of the first electrode layer is exposed in the interlevel dielectric material film through a patterning treatment process to form an interlevel dielectric layer; wherein at least part of the strontium titanate dielectric layer is located in the through hole and in contact with the first electrode layer.

19. The manufacturing method according to claim 18, wherein the step of forming the strontium titanate dielectric layer and the second electrode layer in sequence at the side of the first electrode layer away from the substrate comprises: after forming the interlevel dielectric layer, forming a strontium titanate dielectric material film and a second electrode material film located in the through hole and covering the interlevel dielectric layer in sequence; and removing portions of the strontium titanate dielectric material film and the second electrode material film that cover the interlevel dielectric layer to form the strontium titanate dielectric layer and the second electrode layer.

20. The manufacturing method according to claim 19, wherein after the step of forming the strontium titanate dielectric layer and the second electrode layer in sequence at the side of the first electrode layer away from the substrate, the manufacturing method further comprises: forming an insulation dielectric layer and an etch stop layer in sequence at the sides of the interlevel dielectric layer and the second electrode layer away from the substrate; forming a first via hole and a second via hole formed at intervals, wherein an orthographic projection of the first via hole in the substrate falls within an orthographic projection of an edge area of the first electrode layer on the substrate, and is non-overlapped with orthographic projections of the second electrode layer and the strontium titanate dielectric layer on the substrate, wherein the first via hole penetrates through the insulation dielectric layer, the etch stop layer and the interlevel dielectric layer in sequence and exposes part of the first electrode layer, wherein an orthographic projection of the second via hole falls within an orthographic projection of the through hole in the substrate, the second via hole penetrates through the insulation dielectric layer and the etch stop layer in sequence, and exposes part of the second electrode layer; forming a measurement electrode material film, wherein the measurement electrode material film covers a surface of the etch stop layer away from the insulation dielectric layer and fills the first via hole and the second via hole; and performing patterning treatment on the measurement electrode material film to form a first measurement electrode and a second measurement electrode, wherein each of the first measurement electrode and the second measurement electrode comprises a via hole conduction portion and a measurement conduction portion, the measurement conduction portion of the first measurement electrode and the measurement conduction portion of the second measurement electrode are arranged at intervals and are formed at the side of the etch stop layer away from the substrate, the measurement conduction portion of the first measurement electrode is a measurement electrode material film that fills the first via hole, and the measurement conduction portion of the second measurement electrode is a measurement electrode material film that fills the second via hole.